AJ

ARUN JOSE

SOFTWARE ENGINEER // DENMARK

SIEMENS ENERGY

MSC TU CHEMNITZ

ARUN JOSE

> **“Building useful software, exploring intelligent systems, and connecting engineering with research.”**

WORKSTATION_86 // ENGINEER + RESEARCHER BENCH

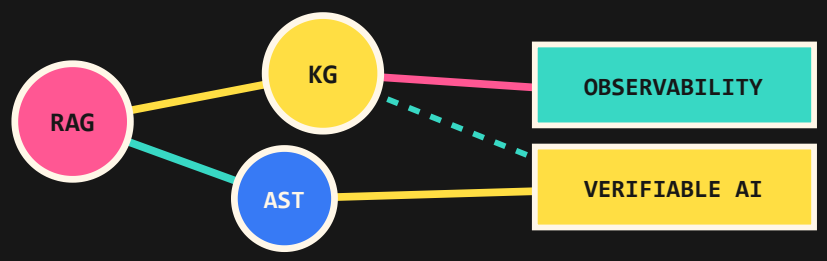
AI + INFRA

PORTRAIT.BMP 8-BIT



ARUN JOSE // PORTRAIT

> KNOWLEDGE_GRAPH_ENGINE.SYS [ONLINE]



- Platform Engineering
- Infrastructure Monitoring
- DevOps & Automation
- Knowledge Graphs
- Software Intelligence
- Applied AI Research

1980S MEMPHIS × NEO-BRUTALIST PORTFOLIO LAUNCH

SWIPE FOR TOUR →

CASE STUDIES // 01-03

FEATURED PROJECTS

Bridging academic applied AI research, enterprise platform reliability, and open-source software intelligence.

PRJ-01 // MASTER'S RESEARCH (TU CHEMNITZ)

THESIS ARTIFACT

HYBRID SEMANTIC FRAMEWORK FOR EXPORT CONTROL COMPLIANCE

Unites dense vector retrieval with symbolic Knowledge Graph traversal to resolve multi-hop regulatory classifications with traceable, explainable evidence chains.

Hybrid RAG | Knowledge Graphs | Semantic Reasoning | NLP | Vector Search

PRJ-02 // PROFESSIONAL EXPERIENCE

ENTERPRISE PLATFORM

INFRASTRUCTURE MONITORING & AUTOMATION

Resilient observability pipelines, high-signal health telemetry, and reproducible CI/CD automation routines designed to eliminate manual operational toil.

Platform Engineering | Observability | DevOps Automation | CI/CD | Reliability

PRJ-03 // PROPOSED OPEN-SOURCE PROJECT

R&D CONCEPT

CODE INTELLIGENCE LAB

Experimental workbench combining AST parsing, dependency property graphs, and hybrid retrieval for verifiable repository-scale impact analysis and developer tools.

Software Intelligence | Code Graphs | AST Analysis | Developer Productivity

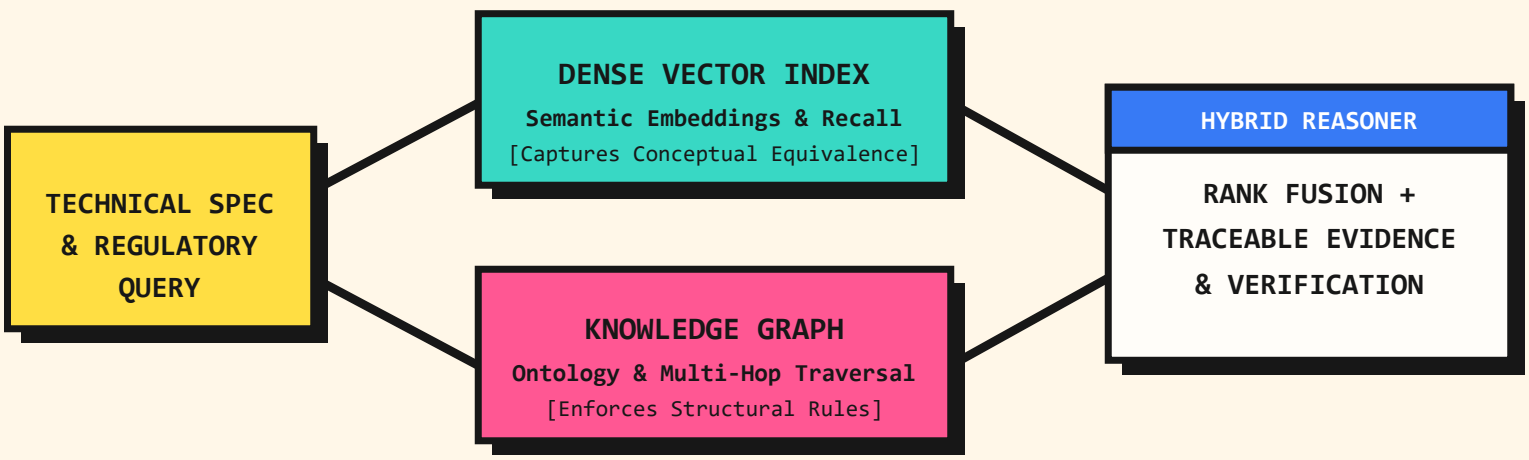
GOVERNANCE NOTE: Public overviews highlight methodology & patterns; confidential internal architectures are strictly omitted.

RESEARCH LAB // TU CHEMNITZ & BEYOND

HYBRID RAG & GRAPHS

Combining neural language models with explicit symbolic structures for verifiable, grounded engineering intelligence.

FIG 02 // HYBRID_SEMANTIC_COMPLIANCE_PIPELINE.SVG — ARCHITECTURE



RES-01 // MASTER'S THESIS

EXPORT CONTROL COMPLIANCE AI

Modeled regulatory hierarchies and technical concepts to assist complex classification with human-in-the-loop verification.

RES-02 // HYBRID RETRIEVAL

VECTOR + SYMBOLIC GRAPH RAG

Fusing dense embeddings, lexical BM25 precision, and graph neighborhood assembly to mitigate hallucinations.

RES-03 // SEMANTIC SYSTEMS

KNOWLEDGE GRAPHS & NLP

Property graphs, entity linking, and domain-adapted NLP for technical corpora and large software repositories.

RES-04 // FUTURE HORIZONS

VERIFIABLE AI ENGINEERING

Neuro-symbolic guardrails, citation-grounded generation, and deterministic policy checks for critical systems.

CAREER CHRONOLOGY & CRAFT

EXPERIENCE & TENETS

International engineering experience across Siemens organizations paired with graduate research in Germany.

STEP 01 // FULL-TIME ROLE • DENMARK

SIEMENS ENERGY — SOFTWARE ENGINEER

2026-PRESENT

Platform engineering, infrastructure automation, system observability, and resilient software delivery.

STEP 02 // WORKING STUDENT • GERMANY

SIEMENS GAMESA — WORKING STUDENT

2023-2026

Infrastructure monitoring workflows, DevOps tooling, and engineering automation alongside MSc studies.

STEP 03 // WORKING STUDENT • GERMANY

SIEMENS AG — WORKING STUDENT

2022

Software development, automated testing workflows, and internal web engineering enhancements.

STEP 04 // FULL-TIME ENGINEERING • INDIA

SIEMENS GAMESA — SOFTWARE ENGINEERING

2018-2021

Enterprise software components, backend & web workflows, and production engineering lifecycle delivery.

ACADEMIC CREDENTIAL // GERMANY

MSC WEB ENGINEERING

Technische Universität Chemnitz

Specialized in Semantic Web, Distributed Systems & Applied AI.

ENGINEERING PHILOSOPHY

OBSERVABILITY & AUTOMATION FIRST

Treat telemetry as a first-class feature, codify repeatable toil, and pair AI research with deterministic guardrails.

* NOTE: Verify exact roles and dates before final publication

MDX BLOG & ARCHITECTURE

WRITING & CONNECT

TECHNICAL NOTEBOOK // 4 MDX WORKING DRAFTS

RSS + MDX

DRAFT // VOL-01

ENGINEERING RELIABLE INFRASTRUCTURE

Observability, Signals & Self-Healing Workflows

DRAFT // VOL-02

KNOWLEDGE GRAPHS FOR CODE INTELLIGENCE

ASTs, Call Hierarchies & Property Graphs

DRAFT // VOL-03

HYBRID RETRIEVAL-AUGMENTED GENERATION

Bridging Vector Search & Symbolic Graphs

DRAFT // VOL-04

SOFTWARE AUTOMATION & PRODUCTIVITY

Designing Low-Friction Engineering Platforms

PORTFOLIO ARCHITECTURE & 1980S MEMPHIS DESIGN TOKENS

#FFF7E8 #FFDE43 #FF5793 #38D8C4 #377AF5

ASTRO 7 • TYPESCRIPT • TAILWIND • MDX

OPEN FOR COLLABORATION

LET'S CONNECT ON LINKEDIN

Platform Engineering • Observability • Knowledge Graphs • Applied AI Research

LOC DENMARK

ORG SIEMENS ENERGY

EDU MSc TU CHEMNITZ

PORTFOLIO LINK IN POST ↓